Computer Systems Modelling: Lent 2025 **Assessment 3**
Instructor: Prof. S. Keshav 10-11 AM, March 20, 2025
Answer all questions Maximum marks 15

Question 1.
Consider the periodic square wave function:

$$f(t) = \begin{cases} 1, & 0 \leq t < 1 \\ 0, & 1 \leq t < 2 \end{cases}$$

where $f(t+2) = f(t)$ for all t. Derive the Fourier series expansion of $f(t)$ as a sum of sines and cosines. **(2 marks)**

Solution:
For a periodic function with period $T_0 = 2$, $\omega_0 = \frac{2\pi}{T_0} = \pi$. So the Fourier series is given by:

$$f(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos k\omega_0 t + b_k \sin k\omega_0 t)$$

$$= a_0 + \sum_{k=1}^{\infty} (a_k \cos k\pi t + b_k \sin k\pi t)$$

First, we calculate a_0:

$$a_0 = \frac{1}{T_0} \int_{T_0} f(t)dt = \frac{1}{2} \int_0^2 f(t)dt = \frac{1}{2} \left[\int_0^1 1 \, dt + \int_1^2 0 \, dt \right] = \frac{1}{2} \cdot 1 = \frac{1}{2}$$

For the cosine coefficients a_k:

$$\begin{aligned} a_k &= \frac{2}{T_0} \int_{T_0} f(t) \cos k\omega_0 t dt \\ &= \frac{2}{2} \int_{T_0} f(t) \cos k\pi t dt \\ &= \int_0^1 1 \cdot \cos(k\pi t) \, dt + \int_1^2 0 \cdot \cos(k\pi t) \, dt \\ &= \frac{1}{k\pi} \sin(k\pi t) \Big|_0^1 \\ &= \frac{1}{k\pi} [\sin(k\pi) - \sin(0)] \\ &= \frac{1}{k\pi} \cdot 0 = 0 \quad \text{for all } k \geq 1 \end{aligned}$$

For the sine coefficients b_k:

$$\begin{aligned}
b_k &= \frac{2}{T_0}\int_{T_0} f(t)\sin k\omega_0 t dt \\
&= \frac{2}{2}\int_0^2 f(t)\sin k\pi t dt \\
&= \int_0^1 1\cdot\sin(k\pi t)\,dt + \int_1^2 0\cdot\sin(k\pi t)\,dt \\
&= -\frac{1}{k\pi}\cos(k\pi t)\Big|_0^1 \\
&= -\frac{1}{k\pi}[\cos(k\pi)-\cos(0)] \\
&= -\frac{1}{k\pi}[(-1)^k - 1] \\
&= \frac{1}{k\pi}[1-(-1)^k]
\end{aligned}$$

Therefore:

$$b_k = \begin{cases} \frac{2}{k\pi}, & \text{for } k \text{ odd} \\ 0, & \text{for } k \text{ even} \end{cases}$$

The Fourier series expansion is:

$$f(t) = \frac{1}{2} + \sum_{k=1}^{\infty}\frac{1}{k\pi}[1-(-1)^k]\sin(k\pi t)$$

$$f(t) = \frac{1}{2} + \sum_{k=1,3,5,\ldots}^{\infty}\frac{2}{k\pi}\sin(k\pi t)$$

Question 2. A website owner measures the number of hits that it receives every second for several years. Treating this as a signal, if you did a Fourier transform, at what frequencies would you expect to find the most energy? Justify your answer. **(1 mark)**

Solution: 1/day, 1/week, 1/year since this is are natural periods at which there is cyclicity.

Question 3. Use the integral rule to find the Laplace transform of the function t^2. **(1 mark)**

Solution: We have

$$t^2 = 2\int t dt$$

So

$$L(t^2) = 2L(\int t dt)$$

But

$$t \leftrightarrow 1/s^2$$

So

$$L(\int t dt) = 1/sL(t) = 1/s\cdot 1/s^2 = 1/s^3$$

$$L(t^2) = 2/s^3$$

Question 4. With dynamic voltage scaling, a CPU increases or decreases its cycle rate as a function of a voltage control signal, allowing server power to be reduced when the CPU load is

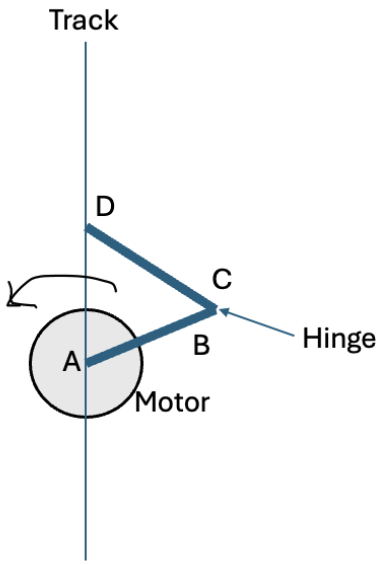


Figure 1: Hinged rotating rods

low. Suppose that the cycle rate $C(t)$ at time t is proportional to the weighted sum of the control voltage at that time $V(t)$ as well as the first derivative of this control voltage, and that this rate is an offset from a fixed base rate C_{min}.

a. Represent the system with an appropriate equation in the time domain (0.5 marks)

b. Is it linear? (0.5 marks)

c. What is the system equation in the Laplace domain? (1 mark)

(Total: 2 marks)

Solution: The time-domain relationship is

$$C(t) = \alpha V(t) + \beta V'(t) + C_{min}u(t)$$

Because of the constant offset term, the system is not linear.
Taking the Laplace transform

$$C(s) = \alpha V(s) + sV(s) + C_{min}/s$$

Question 5. Consider the system in Figure 1. Here, a unit length rigid rod AB is coupled to the shaft of an electrically driven motor at one end A, and the other end B is attached by a hinge to end C of another unit length rigid rod CD so that the rod CD can freely and completely rotate around rod AB. End D of the rod CD is restricted to move vertically along a track as shown: it *must* touch the track at all times. Assume that the two rods are in the same vertical plane at all times and that the motor makes one complete rotation in 2π seconds.

a. If the motion of the rod AB is modelled as a phasor, how would you model the movement of end D of rod CD? **(0.5 marks)**

b. Give the equation of motion of point D *relative to point C* in the form of a complex exponential. Justify your answer. **(1 mark)**

c. Give the equation of motion of point D *relative to point A*. Justify your answer. **(1 mark)**

d. Now, assume that the motor is non-ideal, so cannot be relied to maintain the desired speed of rotation. However, its rotation speed is directly proportional to its input voltage. Describe a control system that can be used to maintain the desired rate of one rotation per 2π seconds. Specifically, write down a table with six rows that identifies the following six features: the plant, the measurement process, the disturbance, the reference input, the feedback, and the control input. **(1.5 marks)**

e. Assume that the motor can be modelled as a first-order system. If, at time 0, a unit voltage is applied to the motor, we find that a rotation speed of .63 radians/s is achieved at time 5 and that the final rate of rotation is 1 radian/s. What is the equation of motion of the motor when given a step input? **(1 mark)**

f. Describe how you could use proportional-mode control to achieve the desired rate of rotation. Identify a problem with the use of proportional mode control in this specific system.**(1 mark)**

Solution:

a. As the sum of two phasors.

b. e^{-jt}: it moves clockwise at the rate of 1 rad/s around point C.

c. 2 sin t (2j sin t is also acceptable), since this is the sum of two phasors with opposite directions of rotation.

Plant	the motor, two rods, the groove, rotation sensor
Measurement process	sensor that measures motor rotation rate
Disturbance	anything that causes the motor to not run at the desired rate
Reference input	1 rad/s
Feedback	actual rotation rate
Control input	control voltage

d.

e. 63% achieved at τ, so equation is $y(t) = 1 - e^{-t/5}$

f. Voltage changes in proportion to error. Since there is need for non-zero control at steady state, there will be some droop.

Question 6. Suppose that the output of a system T, is related to the control variable B, by the equation:

$$2\frac{dT^2}{d^2B} + 3\frac{dT}{dB} + T = B$$

a. What is the transfer function? **(1 mark)**

b. What is the output as a function of time when the system is given a unit step input at time 0? **(1 mark)**

c. Is this system stable? Why or why not? **(1 mark)**

Solution: Taking the Laplace transform of both sides

$$2s^2T(s) + 3sT(s) + T(s) = B(s)$$

$$T(s) = \frac{B(s)}{2s^2 + 3s + 1}$$

So, the transfer function is

$$\frac{1}{2s^2+3s+1}$$
$$= \frac{1}{(2s+1)(s+1)}$$

Given a unit step input, whose Laplace transform is $1/s$, the Laplace transform of the output is

$$= \frac{1}{s(2s+1)(s+1)} = \frac{1}{s} - \frac{2}{s+1/2} + \frac{1}{s+1}$$

and, in the time domain

$$y(t) = 1 - 2e^{-t/2} + e^{-t}$$

Both roots of the transfer function (-1 and -1/2) are in the left half plane, so the system is stable.